

A Decentralized Approach to Sharing Energy Consumption Data of Logistic Buildings

Thomas Wehr¹, Heike Weber², Uwe Veres-Homm² and Andreas Harth^{1,2}

¹Friedrich-Alexander-Universität Erlangen-Nürnberg, Nuremberg, Germany

²Fraunhofer Institute for Integrated Circuits IIS, Nuremberg, Germany

Abstract

Energy consumption data in logistics properties remains fragmented across organizational silos, hindering effective comparison and benchmarking. Existing centralized platforms require stakeholders to relinquish control of their data, which conflicts with privacy requirements and commercial interests. We introduce the concept of Data Rooms – a structured process for stakeholders to negotiate and formalize data sharing agreements before any data is exchanged. Building on this foundation, we present the GRANERGIZE WebApp: a Solid-based application that enables decentralized sharing of building energy data while preserving data sovereignty. The application uses Web Access Control for fine-grained permissions and the Solid Application Interoperability specification for access notifications. In addition to sharing individual building data, the system supports the creation and sharing of aggregated views that only expose computed statistics without revealing underlying buildings. Our implementation demonstrates how the Solid protocol can solve real-world data-sharing issues in the logistics sector, allowing collaboration without centralized data custody.

Keywords

Decentralized Data Sharing, Energy Consumption Data, Logistics Buildings,

1. Introduction

Energy consumption data in logistics properties remains trapped in isolated silos, hindering access and exchange. The logistics ecosystem's complexity – with its diverse stakeholders and varied data formats – creates heterogeneity that prevents comparing and sharing energy data across properties and organizations.

Logistics properties form central hubs of logistics networks and therefore play a decisive role in the provision of logistics services. [1] With rising electricity and gas prices [2] and increasing sustainability requirements through EU taxonomies [3], energy efficiency in logistics real estate is becoming ever more relevant.

Yet substantial barriers to sharing and comparing energy consumption data persist: The stakeholder landscape's complexity – with Facility Managers (FMs), users, investors, brokers, etc. holding different perspectives on energy use – makes it difficult to access all the relevant data for a comprehensive view of energy consumption. Even within organizations, mobilizing consumption data requires extensive manual effort, and the diversity of building types and usage patterns makes meaningful comparison difficult. In addition to these technical and organizational barriers, establishing trust agreement among stakeholders to share sensitive energy data remains a fundamental challenge, as traditional approaches lack transparent mechanisms for negotiating and enforcing data sharing agreements.

Existing centralized platforms fail to address these barriers because they require stakeholders to relinquish data control and adopt singular data models. The Solid protocol offers an alternative: stakeholders retain data sovereignty while enabling controlled sharing, aligning with both GDPR principles and the commercial realities of competitive logistics markets.

The 3rd Solid Symposium Poster Session, co-located with the 4th Solid Symposium, April 30 - May 01 2026, London, UK

✉ thomas.wehr@fau.de (T. Wehr); heike.weber@iis.fraunhofer.de (H. Weber); uwe.veres-homm@iis.fraunhofer.de (U. Veres-Homm); andreas.harth@fau.de (A. Harth)

🆔 0000-0002-0678-5019 (T. Wehr); 0009-0008-3992-6771 (H. Weber); 0009-0000-9216-0615 (U. Veres-Homm); 0000-0002-0702-510X (A. Harth)



© 2026 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

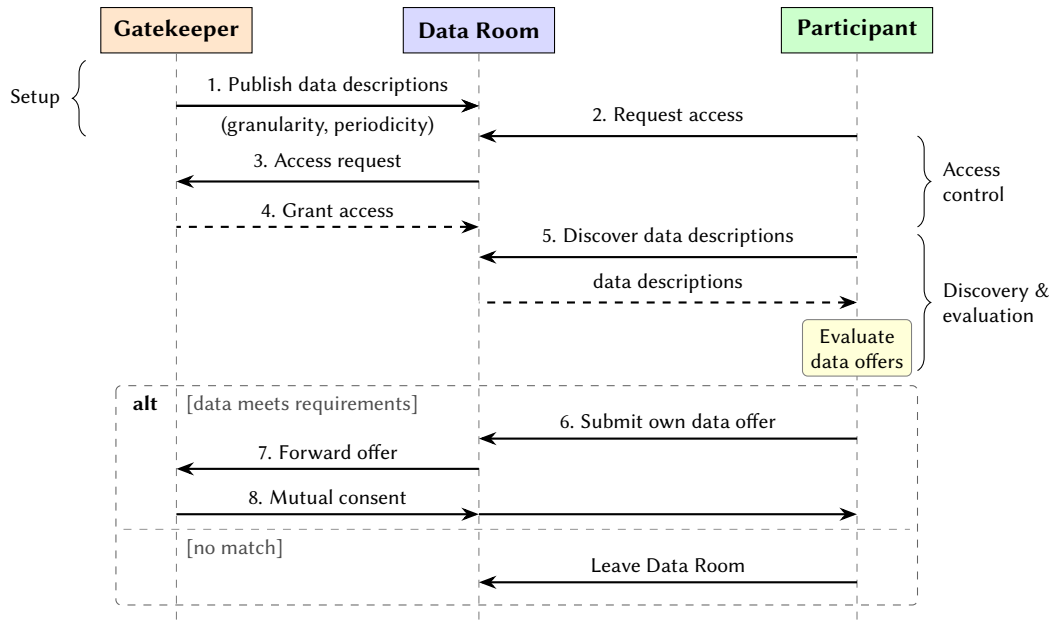


Figure 1: Sequence diagram of the Data Room negotiation process: the gatekeeper publishes data descriptions and controls access; participants discover available data, evaluate offers, and – if requirements are met – submit their own data offers leading to mutual consent.

To address these challenges, the GRANERGIZE project has developed a process that enables stakeholders to negotiate and formalize data sharing agreements, ensuring that consent is established before any data is exchanged. Additionally, we have developed an ontology that provides a unified data model for energy consumption data, as well as a WebApp for visualizing and sharing this data, stored on the user's Solid Pods. The WebApp provides visualizations of the energy consumption data and allows users to define access rights for real or aggregated data via an intuitive user interface. Whereas aggregated data is computed based on the user's preferences and shared as a snapshot without provenance information, individual building data can be shared directly from the owner's Pod with fine-grained access control.

2. Scenario

A central concept in GRANERGIZE is the *Data Room*. Data Rooms are demarcated virtual areas with clearly defined access conditions, controlled by a gatekeeper. Only approved participants are permitted to enter, based on explicit authorization. Within a Data Room, stakeholders can negotiate and formalize data sharing agreements, ensuring that trust and consent are established before any data is exchanged.

We deliberately chose the term Data Room, as these spaces are conceptualized as protected and restricted areas. Access is managed by a gatekeeper – typically the data owner – who determines which participants may enter and under what conditions. The Data Room thus provides a secure environment for stakeholders to discover available data, specify their own data requirements, and negotiate mutually acceptable terms for data exchange. If the data offered meets a participant's needs, they can submit their own data offers in return and, if there is mutual consent, agree to exchange data. Otherwise, participants can leave the Data Room without further interaction.

To illustrate this, consider the following scenario: the facility manager (FM) acts as the owner of a Data Room. Various tenants and brokers are granted access to this room to view descriptions of the data the FM provides, specifying the data's granularity and periodicity. If the data description matches their requirements, they can offer their own data descriptions and, upon agreement, exchange data.

Figure 1 illustrates this negotiation process.

Energy consumption data in logistics properties is relevant for a wide range of stakeholders. While users, owners, and FMs are primarily interested in specific consumption data for reporting and calcula-

Table 1

Overview of stakeholder data exchange pairs involving the facility manager (FM).

Data flow	Data points	Granularity	Periodicity
FM → Broker	Consumption data (for energy performance certificate)	Consumption types*	annually
Broker → FM	User for new properties (as sales support)	Total property	once
FM → Tenant	Consumption data	Consumption types*, as granular as possible	At least monthly, ideally every 15 minutes
	Blind loads	Total property	once
	Peak loads	Consumer types [†]	As detailed as possible
Tenant → FM	Consumption data	Total consumption per user	monthly
	Building specifics [‡]	Number, size, location	annually

Table 2

Overview of stakeholder data exchange pairs between investor and benchmark service provider (BSP). The BSP takes building specifics and consumption data and then provides the calculated consumption per m².

Data flow	Data points	Granularity	Periodicity
Investor → BSP	Building specifics [‡]	Total property	once
BSP → Investor	Consumption data	Consumption types*	annually
	Min/max/median consumption per m ² per property	Consumption types*	annually

tion purposes, developers, financiers, brokers, energy suppliers, and benchmark service providers (BSP) rely on data for property comparisons, accurate energy reporting, and consumption optimization.

To capture these interests and requirements, the project team surveyed each stakeholder group about the implications of energy consumption data for their daily business, the associated pains and gains, and any data requirements they might have from other stakeholders. Table 1 shows an example Data Room owned by a FM, who exchanges consumption data with brokers (e.g., for energy performance certificates) and tenants (e.g., detailed consumption breakdowns including blind and peak loads), while receiving building specifics and tenant-side consumption data in return. Table 2 illustrates a second Data Room in which an investor shares building specifics and consumption data with a BSP, who in turn computes and shares back aggregated consumption per m².

3. Implementation

The scenarios in Sect. 2 impose three key requirements: (1) a data model flexible enough to represent the diverse data points, granularities, and periodicities demanded by different stakeholder pairs; (2) fine-grained access control that allows selective sharing and revocation; and (3) a mechanism for sharing aggregated statistics without exposing underlying building data.

Following the Solid principles of separating identity, data, and application, the WebApp operates as a pure client-side application that communicates directly with Solid Pods without requiring a central backend. All user data resides exclusively in the users' personal Solid Pods. Data sharing is only configured after trust has been established in a Data Room: the implementation assumes that stakeholders have already agreed on the terms of data exchange before any access is granted.

*Electricity, water, and gas.

[†]lighting, intralogistics, etc.

[‡]area, age, shift regime, etc.

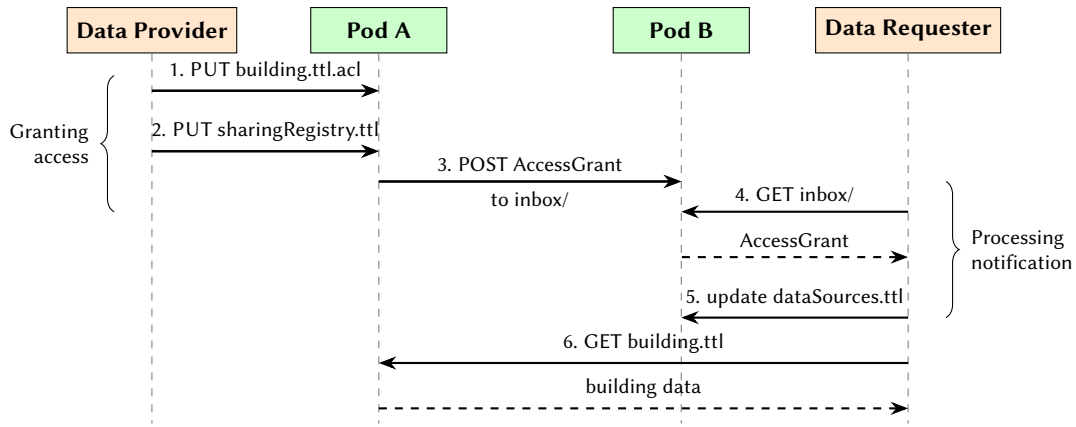


Figure 2: Sequence diagram of the data sharing process: the data provider grants access by updating the ACL and sending a notification; the data requester processes the notification and can then fetch the shared data.

3.1. Data Model

Each building is represented as a Turtle file within the user’s Pod containing static metadata such as type, location, area, and year of construction (see <https://granergize-homer.solidcommunity.net/private/granergize/buildings/2.ttl> for an example). The data model builds upon established ontologies – including RealEstateCore for building classifications and Schema.org for organizational entities – and introduces domain-specific terms via the GRANERGIZE vocabulary (*gra:*). Crucially, a building’s energy measurement data is stored in separate datasets per year, linked from the building file. This separation allows sharing building metadata independently of detailed energy data, supporting the fine-grained access control required by data providers.

3.2. Data Sharing and Access Control

Once stakeholders have reached mutual consent within a Data Room, data sharing is implemented using the Web Access Control (WAC) specification [4]. When the data provider grants access, the application appends an *acl:Authorization* block to the resource’s *.acl* file, specifying the recipient’s WebID and the granted access mode. Building metadata and energy data reside in separate files, each with its own ACL, so the provider can share metadata without exposing detailed measurements.

The sharing workflow follows the Solid Application Interoperability (INTEROP) specification [5] for notifying data requesters about granted access. Figure 2 illustrates the sharing process. The data provider initiates sharing through the web interface by selecting a building and entering the recipient’s WebID. Upon confirmation, the application executes three operations on the data provider’s Pod (steps 1–3 in the figure): it updates the resource’s ACL file to grant read permission, records the sharing action in *sharingRegistry.ttl*, and posts an *interop:AccessGrant* notification to the recipient’s inbox. When the recipient’s application polls the inbox (steps 4–5), it discovers the grant and adds the shared building URI to its local *dataSources.ttl*, after which it can fetch the data directly from the provider’s Pod (step 6).

Revocation removes the recipient’s authorization from the ACL; subsequent access attempts return HTTP 403, prompting automatic cleanup of the recipient’s *dataSources.ttl*.

3.3. Sharing Aggregated Data

While sharing individual building data enables direct collaboration between pairs such as the FM and tenant, the investor–benchmark provider scenario (Table 2) demonstrates a different need: computing aggregated statistics across multiple properties without exposing the individual buildings. The GRANERGIZE WebApp addresses this through aggregated views – snapshots containing only computed values.

```

1 <#view-1770048602886-f5n4sea> a
   ↳ gra:AggregatedViewDefinition ;
2   gra:viewId "view-1770048602886-f5n4sea" ;
3   gra:viewName "Average Gas and Electricity 467" ;
4   gra:aggregationType "average" ;
5   gra:createdAt
   ↳ "2026-02-02T16:10:02.886Z"^^xsd:dateTime ;
6   gra:includesBuilding
   ↳ <../private/granergize/buildings/4.ttl>,
7     <../private/granergize/buildings/2.ttl>,
8     <../private/granergize/buildings/6.ttl> ;
9   gra:includesMetric "gas", "electricity" ;
10  gra:lastComputedAt
   ↳ "2026-02-02T16:10:04.064Z"^^xsd:dateTime .

```

Listing 1: View Definition Example

Figure 3: Create Aggregated View Dialog

An aggregated view is defined by three parameters: a set of source buildings, an aggregation function (sum, average, minimum, or maximum), and a selection of energy metrics. Figure 3 shows the user interface for creating a view, and Listing 1 shows the resulting Turtle definition stored in the owner’s Pod. Note, that the view definition itself – including the source building URIs – is never shared. Only the computed snapshot, containing aggregated values without provenance information, is made accessible to recipients.

When the owner creates or refreshes a view, the application fetches energy data from each source building, applies the selected aggregation function, and stores the result as a separate snapshot file.

Sharing and revoking access to aggregated views follows the same WAC-based pattern as individual buildings. Because snapshots are materialized, shared values reflect the state at the time of last refresh; owners can update by triggering a manual recomputation.

4. Conclusion

Energy consumption data in the logistics sector remains fragmented across organizational boundaries, preventing effective benchmarking and collaboration. Centralized platforms require stakeholders to surrender data control, conflicting with privacy requirements and competitive interests.

This paper introduces the concept of Data Rooms – a structured process for stakeholders to negotiate and formalize data sharing agreements, establishing trust and mutual consent before any data is exchanged. Additionally, we present the GRANERGIZE WebApp, a Solid-based application that addresses these challenges through decentralized data sharing while preserving data sovereignty. The implementation demonstrates how WAC for fine-grained permissions and the INTEROP specification for access notifications can solve real-world data-sharing problems without centralized data custody. The aggregated views feature extends the sharing model to support statistical summaries without exposing individual building details.

A key area for future development is integrating Data Room functionality directly into the WebApp, enabling stakeholders to negotiate and manage data sharing agreements within the application. Additional future work includes integrating automated access request workflows, data integration pipelines, and supporting additional aggregation functions such as weighted averages based on building size. We also plan to evaluate the system’s usability with logistics stakeholders in a field study.

5. Acknowledgements

We acknowledge funding from the German Federal Ministry for Economic Affairs and Energy (FKZ 01IF23286N).

References

- [1] A. Nehm, U. Veres-Homm, C. Kille, Logistikimmobilien in deutschland (2009).
- [2] Federal Statistical Office of Germany (Statistisches Bundesamt), Data on Energy Price Trends: Long-time Series from January 2005 to December 2022, Technical Report, Federal Statistical Office of Germany (Statistisches Bundesamt), 2023.
- [3] European Commission, Eu taxonomy for sustainable activities, 2022. URL: https://finance.ec.europa.eu/sustainable-finance/tools-and-standards/eu-taxonomy-sustainable-activities_en.
- [4] T. Berners-Lee, H. Story, S. Capadisli, Web Access Control, W3C Draft Report, W3C, 2024. <https://solidproject.org/TR/wac/>.
- [5] J. Bingham, E. Prud'hommeaux, E. Pavlik, Solid Application Interoperability, W3C Draft Report, W3C, 2025. <https://solid.github.io/data-interoperability-panel/specification/>.